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Effects of minimum wage changes on labor demand: analysis using job postings data

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ABSTRACT

Using job advertisement data in Japan, we examine the effects of minimum wage (MW) changes on labour demand and how regional differences influence the impact of these changes. Previous research has struggled to fully evaluate the consequences of MW changes because of varying local labour–market conditions and methodological challenges. This study focuses on the demand side by analysing changes in job offers, which better reflect firms' hiring intentions independent of supply shifts. Our findings reveal that MW increases lead to a net increase in job listings, with a rise in jobs above the new MW exceeding the decline in jobs below it, resulting in a total increase of approximately 10–16%. The positive response is more pronounced in low-density, monopsonistic regions, while high-density competitive markets tend to show a reduction in job offers. The negative impacts are particularly noticeable in clerical, manufacturing, logistics, and cleaning jobs, where competition is high. However, professional jobs show a minimal impact. Our results highlight that the effects of MW increases depend on regional labour–market structures and the competitive dynamics of specific occupations – policymakers should explicitly account for such heterogeneity when designing MW policies.

KEYWORDS

Minimum wage; local labour market; job advertisement data; labour demand; monopsonistic markets

JEL CLASSIFICATION

J23; J31; J38; J42; J63

I. Introduction

We used unique data from a Japanese job advertising Company A to analyse the impact of minimum wages (MWs) on firms' job offers. The existing literature has primarily focused on the impact of MWs changes on the number of workers employed by firms. However, we argue that in a labour market such as Japan, where dismissals because of MW increases are not permitted and employment is relatively stable, the impact of MWs manifests more strongly in job offers rather than in employment levels. To the best of our knowledge, except for Kudlyak, Tasci, and Tüzemen (2022), few studies have examined the effects of MWs on job offers as we have done, and our research contributes to the existing literature in this area. Furthermore, we examine how the effect of MWs on the number of job offers differs depending on the state of competition for applicants, which varies across labour markets.

The effect of MWs legislation on employment has been the subject of considerable debate in numerous countries – from an academic standpoint and in the context of economic and social policy. Although the MW system is a significant social issue, previous studies have yielded inconclusive results. Some studies have indicated a negative impact on employment, while others have not found such an effect.

In a perfectly competitive labour market, an increase in the MWs above the equilibrium wage would result in a decline in labour demand and an increase in labour supply. In this case, an increase in the MWs may result in higher wages for some workers. Nevertheless, this will not lead to an improvement in overall welfare because it will also result in job losses. Neumark and Shirley (2022) reported that most (79.2%) of the estimates included in studies that utilized their established criteria indicate a negative value of MW employment elasticity – raising MWs would likely eliminate jobs and reduce economic welfare. Furthermore, roughly half of these estimates were found to be statistically significant. The researchers observed that the influence of MWs on younger workers and those with less education is especially pronounced – an increase in MWs will result in a reduction in their employment.

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Nevertheless, Manning (2021) highlights that the conclusion that MWs have a negative impact on employment is not straightforward, citing several theoretical and analytical challenges. This study examines one such issue – the various problems inherent in a non-competitive labour market with friction. In the context of an imperfectly competitive labour market, Manning (2021) posits that the impact of a MW increase on employment remains a priori unknown. Card and Krueger (1995) and Manning (2003) conclude that imperfectly competitive labour markets do not necessarily reduce labour demand, and if this is the case, MWs could contribute to improved welfare levels.

From the perspective of imperfectly competitive markets, the respective reactions of labour supply and demand to MWs must be differentiated. Portugal and Rute Cardoso (2006), Brochu and Green (2013), and Dube, William Lester, and Reich (2016) analyse the impact of MWs on labour flows rather than on the size of the employees. Although they consider changes in employer and worker behaviour to some extent by distinguishing between leaving employment (outflow) and entering employment (inflow), they could not capture employer and worker behaviour separately. While we cannot observe the response of workers, we can consider the impact of MWs on employers by observing the number of job offers. If MWs reduce the number of job offers, even if labour supply increases, it should have a negative impact on the volume of labour demand. Thus, we consider the number of job offers to be a more significant indicator of the employment market than the volume of employment.[1]

The data utilized in this study, provided by job advertising Company A, allow for categorizing the number of jobs by hourly wage range. This enables an examination of the impact of changes in the MWs on specific wage ranges[2]; how did the number of job openings near the minimum wage and the number of high-paying job openings change after the minimum wage was raised? From the same perspective as our research, Cengiz et al. (2019) analysed the impact of the MWs by focusing on changes in the wage distribution rather than total employment levels. They created bins by dividing the wage distribution for each period into increments of $0.25 and used Difference-in-Difference (DID) to estimate how the employment contained in each bin changed over time before and after the MW change. According to their analysis, employment levels declined significantly among workers whose wages fell below the MWs while exhibiting an increase in employment for those whose wages were slightly above the MWs. However, the influence is constrained to a specific wage range, with no discernible employment shifts observed among those at the upper end of the wage distribution. In the following analysis, we also aggregate job offers by wage bin, similar to them, and estimate the increase or decrease in job offers for each wage bin before and after the MWs change using DID. This allows us to observe which wage levels' job offers are affected and how long the impact persists.

Furthermore, as the competitive state of the labour market varies by region, we attempt to examine how the effects of the MWs differ across regions. Our data can be analysed by region, enabling an examination of the discrepancies in the impact of MWs across disparate labour–market conditions and regions. Significant regional differences exist in wage levels, and part of this is attributable to regional variations in the characteristics of the labour force. However, the offering wages used in this study are for part-time workers. Therefore, worker characteristics have minimum influence on regional differences in wage levels. Nonetheless, regional labour–market characteristics may exert an influence on the impact of changes in the MW. In a competitive labour market with numerous recruiting firms and job seekers, an increase in MWs may potentially lead to a reduction in the number of job offers. As an alternative possibility, if the labour market is characterized by a tight supply–demand balance or if firms have a significant number of unfilled positions, they may attempt to increase

[1]Employers cannot easily fire workers in the Japanese labour market, even if they are part-time workers with fixed-term employment; thus, the sum of the volume of employment and the number of job offers approximates the volume of labour demand.

[2]Clemens and Strain (2021) and Butschek (2022) have found that minimum wage changes have led to employers seeking higher quality workers and point out that changes in the minimum wage may have caused substitution among workers. If this is correct, the degree of impact of the minimum wage should vary by job.

the number of applicants by raising the offered wage during an MW change. Azar, Marinescu, and Steinbaum (2019) observed that the elasticity of firm-level job applications was zero up to the 80th percentile population density region – many workers were in a monopolistic market. In the light of these considerations, this study aims to ascertain whether discrepancies in regional labour–market characteristics influence the impact of MWs. As this phenomenon is similarly observed at the occupational level, the analysis will also be conducted at this level.

Preliminary estimates suggest that the number of jobs offering wages above the MWs will increase following a MW change, while that below the MWs will decline, resulting in an overall net increase in job opportunities. Concurrently, evidence suggests that increases in the MWs result in a concomitant rise in jobs at higher wage levels. However, this effect is mitigated at the higher end of the wage distribution. Analysis of samples segmented by region revealed that alterations in MW regulations tend to reduce employment opportunities in regions characterized by intense competition for applicants. Conversely, such changes have been found to result in an increase in employment opportunities in areas exhibiting low-population density and monopsonic characteristics. Furthermore, an increase in MW can result in a decline in the number of job openings in occupations that face significant competition for applicants. Consequently, our estimates corroborate the theoretical prediction that an increase in the MWs leads to a decrease in job openings in competitive markets and an increase in job openings in monopsonic markets.

The rest of this paper is organized as follows: Section II elaborates the details of the data, methods of analysis, and empirical model. Section III presents the results, and Section IV discusses and concludes the study.

II. Data and empirical models

Data

In this study, we utilize job advertisement data provided by Japanese Company A. The job advertisements published by Company A encompass part- and full-time positions; however, the data we employ are exclusively for part-time positions. Most extant studies have focused on analysing minimum wage-sensitive workers (see Allegretto et al. 2017; Card 1992a, 1992b; Card and Krueger, 1994; Dube et al. 2010; Katz and Krueger 1992; Neumark and Wascher 1992; Neumark et al. 2014). Our analysis aligns with their approaches. A secondary rationale for constraining our analysis to part-time employment is that all recruiters specify the advertised wage for part-time positions as an hourly wage. Moreover, the posted wage is typically implemented at the time of hire.

Our dataset comprises information on approximately 28.26 million job advertisements from October 2018 to February 2024 across the three major economic zones: the Tokyo, Kansai, and Chukyo metropolitan areas. Revisions occur each October, as described in detail in the next section. Accordingly, we focus on annual changes in the MWs from 2019 to 2023. Notably, during the 2020 revision, when the pandemic was at its peak, there was almost no change in the MWs. The dataset includes the publication date, the location of the recruiting establishment, industry, occupation, and the recruiting wage.

Figure 1 compares our dataset from Company A and the contemporaneous number of new vacancies of part-time jobs posted on the Public Employment Security Offices across the country. This assesses the data's alignment with Japanese labour market trends. The trend itself remains relatively consistent until 2019, although there are some notable discrepancies. Company A's postings were slightly below public-agency listings and exhibited month-to-month volatility. To mitigate this, our empirical analysis compares two-month windows before and after each MW revision rather than single months. Both sources saw a sharp decline in 2020 because of the COVID-19 pandemic; however, from 2021 onward, Company A's postings have grown much faster and now exceed those of public agencies. Company A's data serve as a reliable proxy for total Japanese job offers and are being used by the government to publish more timely wage trend statistics (*Nikkei Shinbun*, a major Japanese economics newspaper, 6 April 2021).

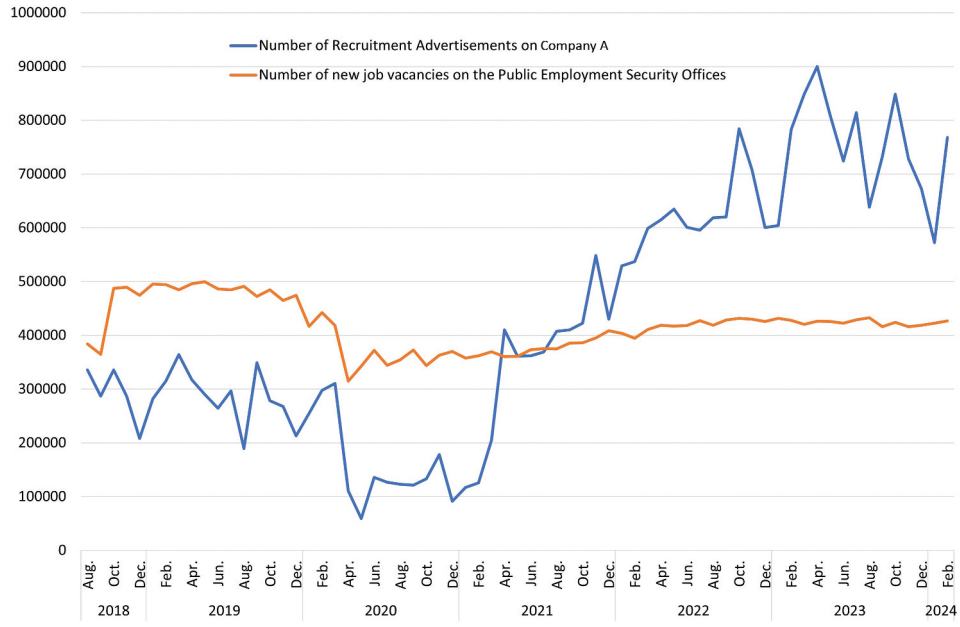


Figure 1. Number of part-time job recruitment on the public employment security offices and Company A.

Figure 2 illustrates the number of job advertisements posted by Company A by region. The number of job advertisements in Tokyo was the highest in all periods, representing 20–40% of the total. The second largest number of job advertisements is observed in Osaka, followed by Aichi and Kanagawa. The Tokyo metropolitan area, which encompasses Tokyo, Chiba, Saitama, and Kanagawa, accounts for approximately half of the total number of advertised positions. Therefore, although the quantity of job advertisements differs across regions, the observed trends exhibit no significant discrepancies.

Figure 3 illustrates the variation in job offer wages (expressed in hourly terms) across different regions. In all periods, the highest offer wages are observed in

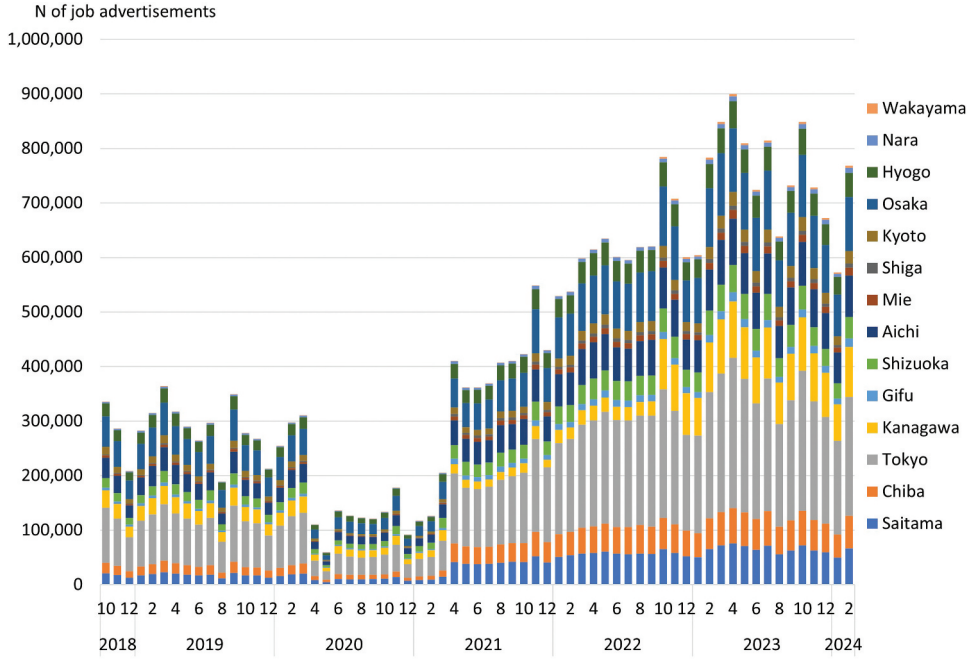


Figure 2. Number of job advertisements posted by Company A (October 2018 ~ February 2024).

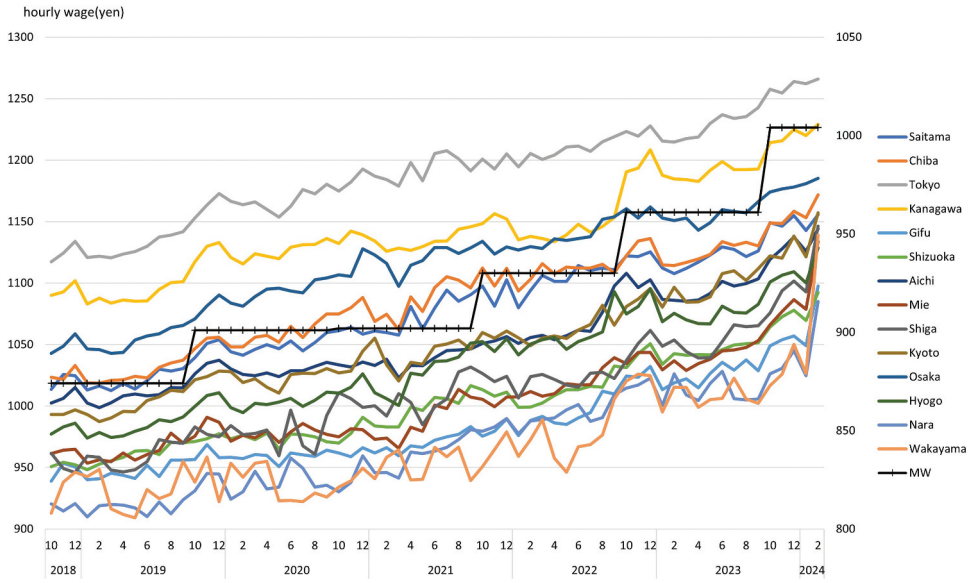


Figure 3. Trends in recruiting wages. MW denotes the nationally weighted average minimum wage across all prefectures, including those outside the Tokyo, Kansai, and Tokai economic zones. The MW axis is displayed on the right.

Tokyo, followed by Kanagawa and Osaka. While the four prefectures in the Tokyo economic zone occupy the top wages, the lowest wages are observed in Wakayama, Mie, and Nara, where the regional population is relatively small. Figure 3 also includes the nationally weighted average MWs trend across all prefectures, including those outside the Tokyo, Kansai, and Tokai zones. Offer wages have risen in all regions in line with MW increases.

Japanese MW system

Table 1 shows the MWs of each prefecture from 2018 to 2023. In Japan, the MW is determined by the Minimum Wage Act, and employers must pay employees an amount greater than the MW. There are two types of MWs: the regional MW, applicable to all employees and employers living in a specific prefecture, and the specific MW for employers and employees in specific industries. Regional and specific MWs are modified simultaneously, and the higher one is enforced. This study uses the regional MW because this type of MW is greater than the specific MW in the three major economic zones that we investigated. This system contrasts with that of the US, where states and localities can set their MWs higher than the federal standard. Returning to the Japanese case, the MW was set

Table 1: Minimum Wage by Prefecture (per hour)

	2018	2019	2020	2021	2022	2023
Tokyo economic zone						
Saitama	898	926	928	956	987	1028
Chiba	895	923	925	953	984	1026
Tokyo	985	1013	1013	1041	1072	1113
Kanagawa	983	1011	1012	1040	1071	1112
Tokai economic zone						
Gifu	825	851	853	880	910	950
Shizuoka	858	885	885	913	944	984
Aichi	898	926	927	955	986	1027
Mie	846	873	874	902	933	973
Kansai economic zone						
Shiga	839	866	868	896	927	967
Kyoto	882	909	909	937	968	1008
Osaka	936	964	964	992	1023	1064
Hyogo	871	899	900	928	960	1001
Nara	811	837	838	866	896	936
Wakayama	803	830	831	859	889	929

Currency:Yen

up according to the following process every year. First, the Central Minimum Wage Council ranks all prefectures from A to D, determines the guide level of MWs for the next year, and proposes it to each prefectural local MW council. This process lasts until the end of July; by late August, local MW councils set regional rates based on the guide level, and after the objection period, the new rates take effect around October 1.

Major newspapers such as *Asahi Shimbun* and *Nikkei Shimbun* reported the Central Minimum Wage Council's guide levels in late July and the local councils' decisions in early August – the public was aware of the impending October increases. Figure 3 shows that recruiting wages began rising before the October 1 implementation, a trend especially pronounced in 2019, 2022, and 2023 following substantial minimum-wage hikes.

Effect of MW changes on the number of job offers: a conceptual framework

To clarify the identification strategy used in our empirical analysis, we first provide a conceptual explanation of how changes in MWs may affect job offers. The following discussion outlines the key idea behind our approach using the structure of Japan's MW system and the heterogeneity in wage distributions, particularly the differences between jobs close to the MW and those further above it.

In accordance with the Japanese MW system, changes in MWs are implemented concurrently throughout the country, although the magnitude of the increase differs across regions. Thus, a comparison between regions cannot be made where there has been a change in the MWs and regions where there has been no change. However, as our data provide information on recruiting wages for jobs on a precise yen-by-yen basis, we can compare the impact of jobs situated close to the MWs with those situated further away. By employing these disparities in wage levels, we can ascertain the influence of the MW on the number of openings at each wage range by utilizing the DID estimation methodology proposed by Cengiz et al. (2019).

Figure 4 is a theoretical graph showing the effect of a change in MWs on the number of job offers. The hypothetical frequency distribution of wages offered before and after the change in MWs is shown by the red and blue dashed lines in this figure. When the changed MWs is set to MW in the figure, all job offers in the red-dashed area should disappear if the labour market is perfectly competitive, as previously legal jobs in this area have become illegal. Conversely, if the labour market is a monopsony, jobs where marginal productivity exceeds the MW may shift from the red-dashed area to the blue-dashed area. In this case, the impact of MW on the quantity of jobs can be calculated as the sum of the jobs that remain because of marginal productivity

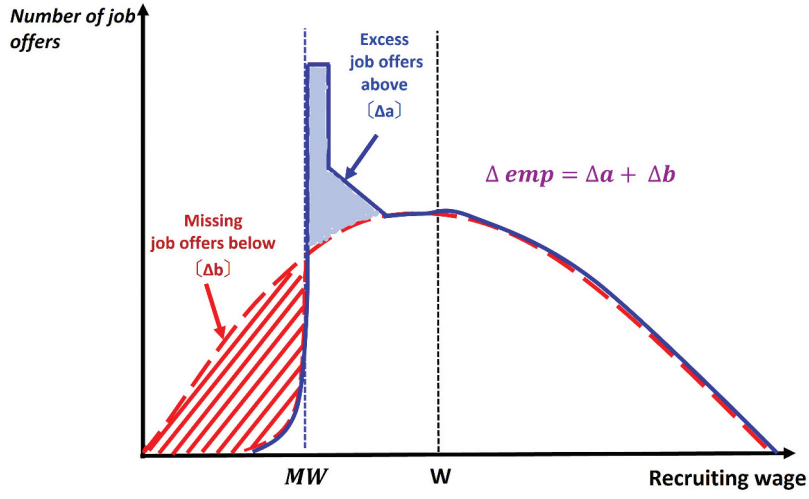


Figure 4. The impact of minimum wages on the frequency distribution of the recruiting wage.

exceeding the MW (Δa) and the jobs that are displaced because of marginal productivity falling below the MW (Δb)[3].

The impact of the MW change on the number of job offers is anticipated to decline at a specific threshold (W in Figure 4) because the jobs in these areas existed prior to the MW change and possess considerably higher marginal productivity. In such a case, a shift in the MW may not impact the volume of job offers or wages for a particular occupation if there are labour–market frictions that diminish the substitutability between high- and low-wage occupations or create a mismatch between job seekers and job offers. As the scope of our analysis is limited to data on part-time employment, job seekers and employers are unlikely to search different segments of the labour market because of high substitutability between occupations. Nevertheless, we also examine the extent to which MW effects affect wage levels.

Furthermore, we are interested in evaluating whether regional labour–market conditions influence the impact of a change in MWs. Competitive conditions in the labour market may exhibit regional variation, which could give rise to disparate MW impacts. Competitive conditions may be more severe in regions with a greater number of firms and workers. Consequently, an increase in MW may result in a reduction in the job offer rate. Conversely, in a tight labour market or one with numerous unfilled vacancies, firms may respond to an increase in MWs by raising wages to incentivize job seekers. Therefore, the size of a region's population and the accumulation of human capital may influence the availability of MWs and the volume of job offers. Additionally, we look at differences in the effect of MWs by job type, as we expect competition to vary by job type even within the same region; firms' response to an MW increase and job seekers' entry behaviours are likely to differ by the nature of the job type.

Empirical model

Base model

As explained in the previous section, we cannot compare regions that had a change in the MWs with those that did not, as Cengiz et al. (2019) have done. Instead, as we can capture jobs on a yen-by-yen basis, we can observe the difference between the impact on jobs near to and far from, respectively, the MW level before and after the MW change. We verify the impact of MWs using the DID approach employed by Cengiz et al. (2019). The specific estimation formula is as follows.

$$\frac{E_{sjot}}{N_{st}} = \sum_{\tau=-5}^{4} \sum_{k=-3}^{12} \alpha_{\tau k} I_{sjot}^{\tau k} + \mu_{sj} + \rho_{jt} + \varphi_o + u_{sjt} \tag{1}$$

Here, the treatment dummy variable $I_{sjot}^{\tau k}$ equals 1 if the recruiting wage bin j is between k and $k+50$ yen corresponding to the new MW after τ months since the MW was increased in month t. For example, $\tau = 0$ is the first month after the MW was increased, and $\tau = -1$ is the month prior. We computed the influence on recruitment for a 25-yen (approximately a quarter of the dollar) interval recruiting wage bins.

The treatment variable I is controlled by wage bin, location (municipality), and time. For example, k = 0 refers to the two wage bins between the MW and MW + 50 yen, whereas k = −1 refers to the two MW bins, MW-1 yen to MW-50 yen. The MW depends on the prefecture in which the municipality s is located. Therefore, a certain wage level is determined as k = 0 in one prefecture; however, even if the wage level is the same, it is classified in the other wage bin in the other prefecture. Thus, a certain wage level is affected by the change in the MW in one prefecture, but the same wage level is not influenced when the MW changes countrywide.

Additionally, this study controls for prefecture-by-wage-bin, μ_{sj}, period-by-wage-bin effects, ρ_{jt}, and occupation dummy, φ_o. u_{sjt}, is the error term. A parallel trend is assumed in the DID analysis. If the MW does not change, the distribution of recruiting wages for the treatment and control groups should change parallelly. Although this assumption cannot be confirmed directly, there is no major difference among prefectures regarding the change in recruiting wage before the change in the MW, as shown in Figure 3, nor in the coefficient of variation.

[3]Job opening below the MW continues even after the change in Figure 4; however, this reflects illegal behaviour by some firms or miscalculations.

For the left-hand side of Equation (1), we use the number of job offers in each wage bin, normalized by the total number of job offers in the municipality and month. Specifically, E_{sjot} is the number of job offers in the municipality s in month t in the occupation o in recruiting wage j, while N_{st} is the total number of job offers of municipality s in month t.

We can calculate the change in the number of job offers because of an MW increase using the estimate coefficient, $\alpha_{\tau k}$, from Equation (1). For the change in the quantity of job offers that has recruiting wages below the MW with a lower limit of 'MW–100 yen' from the month when the MW changed until τ months, it is $\sum_{k=-2}^{-1}\alpha_{\tau k}-\sum_{k=-2}^{-1}\alpha_{-1k}$. As Equation (1) is a DID estimation model, the difference is equivalent to the change excluding the change in the counterfactual distribution. Similarly, the change in the quantity of job offers between the recruiting wage, W, and the minimum wage, MW, is calculated as $\sum_{k=0}^{W-MW}\alpha_{\tau k}-\sum_{k=0}^{W-MW}\alpha_{-1k}$. W = MW + 50 yen serves as the baseline estimate.

Furthermore, the quantity of job offers above the MW is determined by $\Delta a_{\tau}=\frac{\sum_{k=0}^{W-MW}\alpha_{\tau k}-\sum_{k=0}^{W-MW}\alpha_{-1k}}{EPOP_{-1}}$, whereas job offers below the MW are determined by $\Delta b_{\tau}=\frac{\sum_{k=-2}^{-1}\alpha_{\tau k}-\sum_{k=-2}^{-1}\alpha_{-1k}}{EPOP_{-1}}$. $EPOP_{-1}$ is the ratio of job offers to the population and the sample average taken from a 7-month period before the estimations. We divide by $EPOP_{-1}$ to normalize the untreated total job offers. Δa+Δb is the change in total job offers after an increase in the MW. To calculate Δa_{τ} and Δb_{τ}, we determine that July and August of each year as the period before the MW changes and September and October as the periods after the change. The new MW takes effect in October, but the regional MWs are announced in early August, as mentioned above, and the recruiting wages react.

Exploring the regional differences

To address the different impacts of the change in the MWs on the regional labour market, we hypothesize that many company are offering jobs, and many seek jobs in regions with high-population density. Thus, the local labour market is relatively competitive; contrarily, the labour market tends to be monopsony in the low-population density region. If the local labour market in the low-population density region is nearly monopsony, company in those areas rent or afford to increase their wages when the MW increases. Conversely, in the high-population density region, the local labour market is competitive, job offers below the new MW disappear, and the number of job offers decreases when the MW changes. Therefore, we add the cross-term between the treatment variable I and the population density in the Equation (1) and estimate the following:

$$\frac{E_{sjot}}{N_{st}}=\sum_{\tau=-5}^{4}\sum_{k=-3}^{12}\alpha_{\tau k}I_{sjot}^{\tau k}+\sum_{\tau=-5}^{4}\sum_{k=-3}^{12}\beta_{\tau k}I_{sjot}^{\tau k}\times pop+\mu_{sj}+\rho_{jt}+\varphi_{o}+u_{sjt} \tag{2}$$

pop represents the municipal population density in 2019, which is the first year of our estimation period. In areas with high-population density, the labour market is more competitive, and company are less able to raise wages relative to those in municipalities with lower population density when the MW increases. Therefore, we expect β to be negative. If the MW is raised, the increase in the offered wage should be even larger, and the effect of the MW may vary depending on the supply–demand balance in the local labour market. Furthermore, this study conducts analyses by occupation. The impact of the MW varies by occupation, even within the same region.

Table 2 presents descriptive statistics. The average wages and MWs of job postings increase across all three economic zones. The average wage is highest in the Tokyo economic zone, followed by the Kansai economic zone. Regarding the population density in Table 2, the population density in the Chukyo economic zone is approximately half of that in the Tokyo and Kansai economic zones. Additionally, the standard deviation of the Tokyo economic zone is the largest. The average number of job postings categorized by wage bin, year, and month is approximately 1000 to 3000 across three economic zones.

Table 2. Descriptive statistics.

	Tokyo Economic Zone					Kansai Economic Zone					Chukyo Economic Zone				
Variable	Obs	Mean	Std. dev.	Min	Max	Obs	Mean	Std. dev.	Min	Max	Obs	Mean	Std. dev.	Min	Max
wage(yen)															
2019	196,981	1123.037	192.000	762	4000	119,723	1055.827	180.237	762	4000	85,459	1030.061	169.655	770	4000
2020	211,310	1132.502	190.916	792	4000	117,655	1071.407	184.305	800	3900	88,332	1035.709	164.526	800	3800
2021	238,485	1146.440	200.145	793	4000	129,892	1085.140	187.136	800	4000	106,857	1044.816	168.200	825	3800
2022	293,881	1170.905	192.874	821	4000	168,097	1113.363	182.913	821	3600	131,461	1073.017	173.441	825	3785
2023	302,759	1217.912	192.881	853	3700	173,045	1161.265	189.036	880	4000	126,111	1113.278	172.970	872	3800
Population density(peopl e/km^2)	1,243,416	9977.788	5582.859	360.2345	22,252.73	708,412	8449.530	4678.912	354.3391	20,284.72	538,220	3709.117	2864.901	318.9407	10,098.18
*The number of Job Postings by Wage Bin, Year, and Month**															
	MW + up to 25 yen bin		MW + 25–50 yen bin			MW + up to 25 yen bin		MW + 25–50 yen bin			MW + up to 25 yen bin		MW + 25–50 yen bin		
year	2019	2023	2019	2023		2019	2023	2019	2023		2019	2023	2019	2023	
May	3347	3497	2342	1977		2375	2161	1922	1789		2047	2308	852	595	
Jun.	1718	1960	1724	1596		1503	1081	1549	1553		1396	1590	588	489	
Jul.	1586	1853	1890	1625		1470	815	1753	1632		1528	1600	713	532	
Aug.	1536	1745	1862	1537		1471	756	1701	1515		1396	1503	452	474	
Sep.	1681	1904	1889	1636		1509	799	1685	1609		1459	1597	676	524	
Oct.	3648	4371	2155	2246		2478	2770	1881	2131		2067	2678	828	781	
Nov.	3620	4114	2118	2100		2432	2508	1770	1905		2064	2430	819	689	
Dec.	3402	3797	2000	1915		2236	2249	1618	1687		1922	2255	786	613	
Jan.(20/24)	3636	4123	2284	2107		2424	2480	1787	1877		2089	2283	788	644	
Feb.	3789	3964	2342	2133		2572	2283	1841	1727		2112	2240	865	664	
	MW − up to 25 yen bin		MW − 25–50 yen bin			MW − up to 25 yen bin		MW − 25–50 yen bin			MW − up to 25 yen bin		MW − 25–50 yen bin		
year	2019	2023	2019	2023		2019	2023	2019	2023		2019	2023	2019	2023	
May	2504	1830	1561	2738		1635	1260	739	1542		823	634	962	1646	
Jun.	2712	2194	1740	3289		1855	1637	842	1874		966	821	1025	2003	
Jul.	3113	2624	2104	3733		2158	1971	1052	2189		1170	967	1245	2281	
Aug.	3111	2458	2157	3648		2103	1840	1068	2186		931	880	1150	2206	
Sep.	3163	2627	2200	3985		2248	1991	1092	2312		1171	976	1293	2449	
Oct.	312	346	103	197		110	166	38	152		21	84	74	197	
Nov.	312	255	106	106		94	79	54	55		26	24	63	56	
Dec.	263	211	97	70		49	58	54	47		14	23	41	45	
Jan.(20/24)	288	223	95	94		72	82	50	38		12	24	62	58	
Feb.	286	229	89	94		91	78	56	37		12	14	82	73	

*Data for 2020, 2021, and 2022 were also estimated, but only 2019 and 2023 are shown in the table. Similarly, while we have wage bins that go beyond the ±50 yen range of the minimum wage (MW), we only present the bins for up to 25 yen and 25–50 yen above and below the MW.



Figure 5. The impact of the minimum wage on the three major economic zones. Bars show the estimated values of job offers by wage bins relative to the MW (in ¥50 increments): Δb for bins below the MW and Δa for bins above the MW.

III. Results

Effects on the three major economic zones

We estimate Equation (1) using data from the three major economic zones and calculate Δa_τ and Δb_τ, which represent the quantity of job offers above and below the new MW, respectively. Figure 5 indicates the Δa_τ and Δb_τ at increments of 50 yen below and above the new MW for the three major economic zones: Tokyo, Kansai (the second largest), and Chukyo (the third largest) economic zones. The total change in the number of job offers above and below the MW by 50 yen ($\Delta a_\tau + \Delta b_\tau$) corresponds to approximately a 2%. However, Figure 5 shows that the impact of MWs extends to the upper wage ranges but diminishes in the higher ranges. This result is consistent with those of Cengiz et al. (2019) and Lee (1999). When we sum the effects of the increase in the number of job offers up to the wage category 250 yen above the MW, the total change in the number of job offers in the net wage range (from 50 yen below to 250 yen



Figure 6. Impact of the change in minimum wage on job offers by year. Bars show the estimated values of job offers by wage bins relative to the MW (in ¥50 increments): Δb for bins below the MW and Δa for bins above the MW.

above the MW) is approximately a 21% increase. At the few wage ranges on the left end in Figure 5, some increases in the job offers exist, but this is caused by slow action. The reaction of job offers whose recruiting wage is below the MW appears later than that of job offers whose recruiting wage is above the MW.[4]

As indicated in Table 1, the magnitude of changes in the MW varies across years. Therefore, Figure 6 presents the estimation results of Equation (1) by year. From each tick mark on the horizontal axis, the estimated results for each year from 2019 (light blue-green) to 2023 (dark blue) consecutively move to the right. Figure 6 shows that in the wage categories 50 yen above the MW, as well as in those further above, the bar heights are higher in 2019 (light blue-green) and 2023 (dark blue), when the increase in the MW is larger. Conversely, in 2020, when there was almost no revision to the MW (see Table 1), there was hardly any decrease in job offers below the MW by 50 yen, nor any increase in job offers above the MW by 50 yen, as shown by the bars in Figure 6.

Furthermore, Figure 7 presents the results by the three major economic zones and by year. The results are similar across the three zones, but the magnitude of the effect and the range of impact from the change in MW differ. According to Figure 7, in the Tokyo economic zone, the increase in the MW led to a rise in job offers above the MW by 50 yen, particularly in 2019 and 2023, and a decline in offers below the MW by 50 yen, especially in 2023. The total change in the number of jobs 50 yen above and below the MW, increased by approximately 12.1% and 7.6%, respectively, in 2019 and 2023. The impact of the change in MW extended to wages approximately up to 100 yen above the MW. Figure 7 omits wage categories that are more than 250 yen above the MW.

The Kansai economic zone shows similar results. The total change in the number of jobs 50 yen above and below the MW increased by approximately 6.0% and 9.1%, respectively, in 2019 and 2023. The impact has been more widespread in the Kansai economic zone than in the Tokyo economic zone, reaching wage categories approximately 150

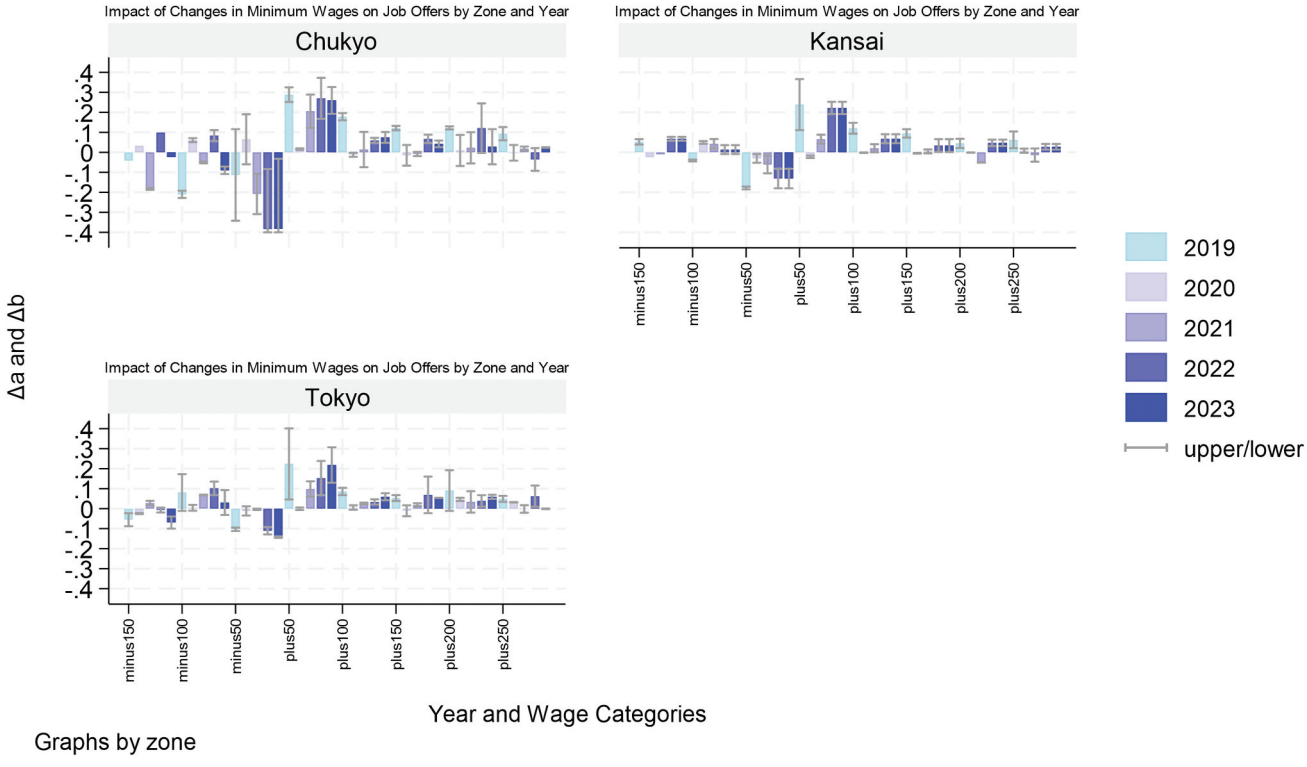


Figure 7. Impact of changes in minimum wages on job offers by zone and year. The interval from the 'plus50' mark to just before the next 'plus100' mark represents the wage bin at the minimum wage plus 50 yen.

[4]As explained earlier, September and October are estimated as the periods after the MW change, but the law came into effect in October. Consequently, some job offers below the MW remain.

yen above the MW. Figure 7 also illustrates a net growth in job offers in the Chukyo zone in 2019. Although the decline in job offers below the MW by 50 yen in 2022 and 2023 is significant, and the total changes in job offers above and below the MW by 50 yen are negative in these years, the number of job offers above the MW by 100 yen and 150 yen has increased in the Chukyo zone, similar to other economic zones.

While the overall effects of MW increases appear broadly similar across the three major economic zones, these aggregate outcomes may conceal significant heterogeneity in labour–market conditions within and between regions. In particular, variations in the degree of labour–market competition and occupational composition could produce region – specific responses to changes in the MW. In the next section, we investigate these potential sources of variation by examining the intra–regional differences in population density and monopsony power.

Analysing the different impacts by local labour market

Figure 8 indicates the estimated coefficient β_{0k} ($\tau = 0$), which is the coefficient of the cross-term between the treatment variable I and population density in Equation (2), for the Tokyo, Kansai, and Chukyo economic zones. $\tau = 0$ represents the first month after the MW increase – October. In Figure 8, the estimated coefficients for wage ranges above the MW by 50 yen or more are almost negative, as expected – labor markets in higher population density areas are more competitive and cannot increase job offers when the MW is raised. The negative effects were particularly pronounced in the Chukyo economic zone and the Tokyo economic zone in 2019. Comparing Chukyo (the third-largest economic zone) and Tokyo in 2019, the negative magnitude is larger in Chukyo. In 2023, these effects are even more pronounced. When comparing Chukyo and Tokyo in 2023, Chukyo shows a more substantial negative impact – even at similar levels of population density, competition tends to intensify in more rural areas within the three major economic zones. Additionally, since the 2021 revisions, MW increases have been higher in rural areas compared to urban centres.

Different impacts among occupations

In Sections 3.1 and 3.2, we demonstrate the impacts of rising MWs on the entire economy within the

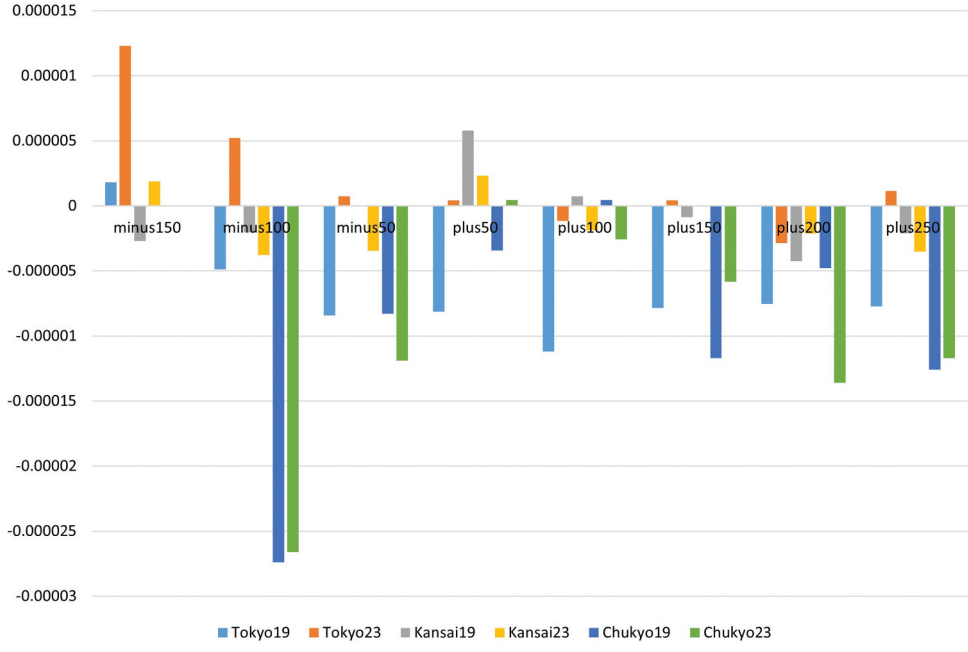


Figure 8. The impact of population density on the effect.
Note: The '19' or '23' following the names of the economic zones indicates that the estimation results are based on data from 2019 or 2023, respectively.

three major economic zones. However, the average recruiting wage, labour shortages, and labour–market competitiveness differ across occupations. Therefore, we conduct an estimation similar to that in Section 3.2, this time by occupation. Using data from the Tokyo economic zone in 2023, we estimate Equation (2) by major occupational categories.

Figure 9 indicates Δa_τ and Δb_τ at increments of 50 yen below and above the new MW by occupation. This trend is the same as the results for the entire economy in Section 3.1, but the volumes vary among occupations. For clerical jobs, the decrease in the number of job offers 50 yen below the MW exceeds the increase 50 yen above the MW. Additionally, for professional jobs, neither the decrease in the number of job offers in the wage range from the MW to 50 yen below nor the increase in those of MW to 50 yen above is significant.

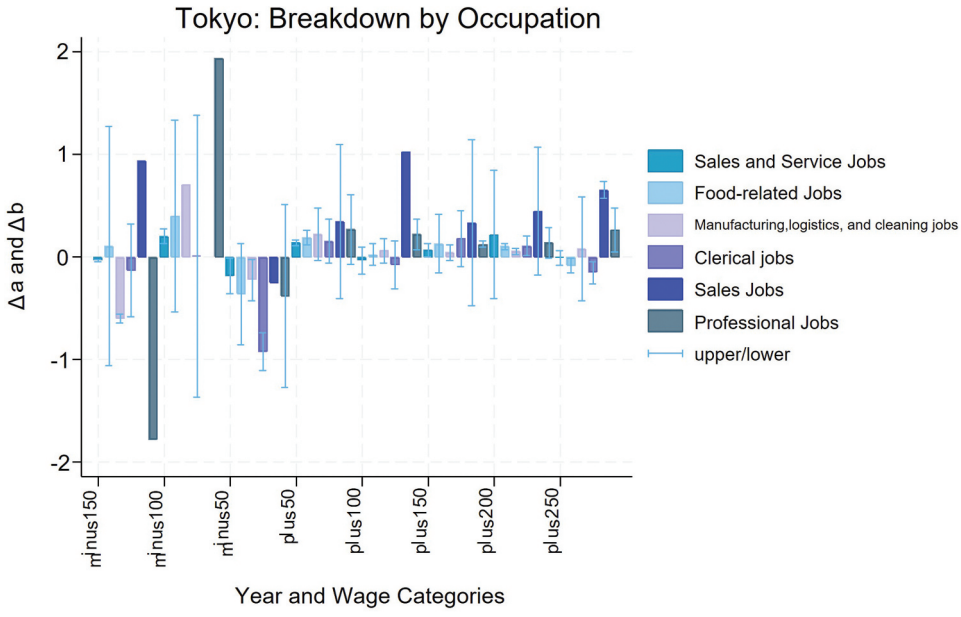


Figure 9. Tokyo: breakdown by occupation.

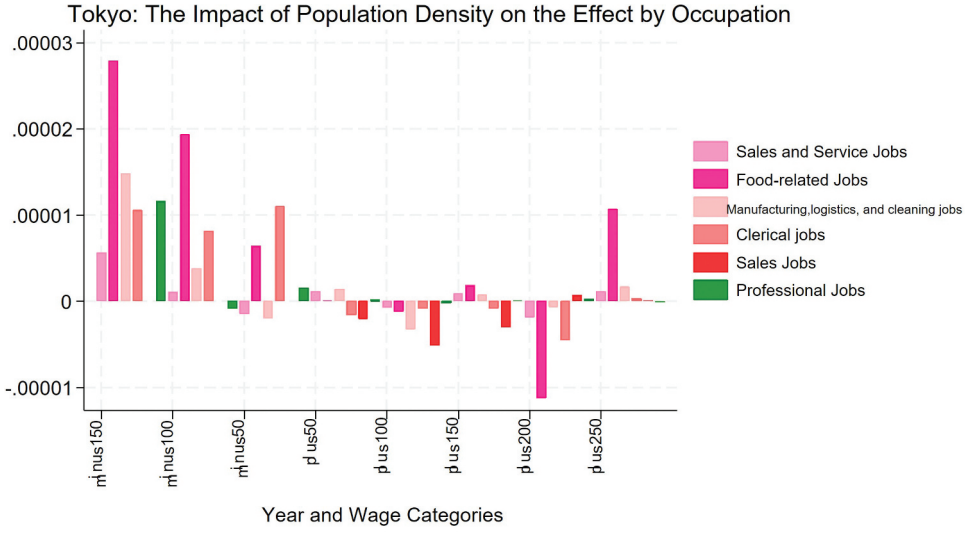


Figure 10. Tokyo: the impact of population density on the effect by Occupation.

Figure 10 shows the estimated coefficient of the cross-term between the treatment variable I and population density by occupation. As expected, the impact of population density on the effects of changes in the MWs is greater for clerical jobs and jobs in manufacturing, logistics, and cleaning, where competition is particularly intense. Nevertheless, the results for food-related jobs do not align with expectations. Considering the post-COVID impact – the recovery in demand for restaurants in densely populated areas and the resulting increase in job offers – we also conducted an estimation using 2019 data. Although the results are not presented, the cross-term for food-related jobs is negative. Similar to the 2023 results, the negative effect of population density is more pronounced in food-related jobs, clerical jobs, and jobs in manufacturing, logistics, and cleaning, while it is less significant in professional jobs.

IV. Discussions and conclusions

We consider the impact of MW on labour demand by observing the number of job listings. A growing body of research on changes in MW has been conducted. Nonetheless, the impact of MW is elusive because it depends on labour–market conditions and has various theoretical and analytical difficulties. Therefore, we treat the number of job listings as a superior indicator of labour demand, as it captures employers' hiring intentions independently of supply shifts, enabling a clear distinction between demand- and supply-side responses to MW adjustments. Simultaneously, this study analyzes how different local labour–market conditions affect the impact of a change in the MW. Previous studies divide the regions based on how high or low the MW is and do not investigate the factors for disparities. Additionally, based on prior findings, while a MW change affects employment with wages close to and above the MW, regional differences have not been examined.

First, the increase in the number of jobs offered above the MW by 50 yen exceeds the decrease below the MW by 50 yen. Total changes are approximately 10–16% of monthly number of job offers in the prior estimation period. The change in the MW influences the upper wage ranges; however, the influence diminishes as the wage range increases. Second, we use population density as a measure of competitiveness in the local labour market. We empirically demonstrated that the high-population density and highly competitive local labour market decrease job offers. Conversely, low-population density and a monopsony labour market increase job offers when the MW increases. Consequently, company such as chain stores are the sole place of work in low-population density regions, and the wage level may stick to the MW. Finally, we examine the effect of the MW rise by occupation. As expected, in occupations where competition is likely to be intense, the following theoretical prediction holds: 'In competitive markets, raising the MW reduces job offers, while in monopsony markets, it increases them'.

Labour–market heterogeneity is not unique to Japan. Numerous studies have documented monopsony power; consequently, developed economies are characterized by labour–market heterogeneity. Therefore, the insights from this study have relevance – and offer guidance – for policymakers and researchers in other advanced countries.

This study has some limitations. As our dataset does not include information on the recruiting company, even if the amount of recruitment remains unchanged when the MW changes, we cannot determine whether the reaction resulting from the same company or the observation through the recruiting company are exchanged.[5] Additionally, with more information about the company, we can analyse why the changes in the MW affect the wages of those with pay already above the MW or whether the decrease in the effect of the change over time is a part of the adjustment process. Moreover, this study does not analyse the long-term effects, considering the change in the supply behaviour of individuals, including migration. Although we demonstrated the different impacts of MWs across occupations, a more detailed analysis of the supply side would also enable us to examine how the impact of MWs differs by gender, as occupational choices often

[5]For example, Dustmann et al. (2022) investigate the reallocation.

vary between men and women. These are implications for further research in the future.

In Japan, the current administration has committed to increasing the MWs to at least ¥1,500 by 2030, which is equivalent to a 7% annual increase. Since the end of the pandemic in 2021, the MWs has increased annually by 3.7%. Leaving aside the debate over what the optimal level should be, according to our findings, even if the MWs increase, the likelihood of a net decrease in job offers is low. Instead, the effect of an average increase in wages offered in job offers is significant. However, owing to the heterogeneity of labour markets across regions, the effects of MWs increases may vary. Our findings suggest that policymakers should place greater emphasis on labour–market heterogeneity when setting regional MWs, rather than seeking to avoid being designated as the lowest MW region nationwide or to prevent setting a wage lower than that of neighbouring regions, and rather than responding to governors' concerns about their public image among local residents.

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